

MODEL TRAINING REPORT

Electronic Nose Coffee Classification System

Model: Long Short-Term Memory (LSTM)

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TRAINING SUMMARY

Training Accuracy	Validation Accuracy	Training Time
97.9%	97.9%	3.0min

MODEL CONFIGURATION

Model Type	Long Short-Term Memory (LSTM)
Epochs	50
Data Split	80% Training / 20% Validation
Input Shape	8 sensors x 300 timesteps
Architecture	Recurrent Neural Network (LSTM)

GAS SENSORS USED (8 Sensors)

TGS2600	MQ135	MQ3	MQ6
MQ7	TGS2602	TGS2611	TGS2620

POWER CONSUMPTION DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	7.45 W	Avg CPU Temp	48.8 °C
Peak Total Power	7.97 W	Peak CPU Temp	50.0 °C
Avg CPU+GPU Power	2.79 W	Energy Used	1310.2 J (0.3639 Wh)
Peak CPU+GPU Power	3.22 W	Training Duration	3.0 min

MODEL EVALUATION RESULTS

Training Data — Accuracy: 97.92%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9694	0.9894
Recall	0.9896	0.9688
F1-Score	0.9794	0.9789
Support (n samples)	96	96

Validation Data — Accuracy: 97.92%

Metric	High Grade (Arabica)	Low Grade (Robusta)
Precision	0.9600	1.0000
Recall	1.0000	0.9583
F1-Score	0.9796	0.9787
Support (n samples)	24	24

Confusion Matrix (Validation Data)

	Predicted: High	Predicted: Low
Actual: High	24	0
Actual: Low	1	23

TP = True Positive (High predicted as High)

FP = False Positive (Low predicted as High)

TN = True Negative (Low predicted as Low)

FN = False Negative (High predicted as Low)

LSTM Summary -- 50 Epochs

Train Loss	0.069808
Validation Loss	0.062094
Total Epochs	50

Accuracy Curve (Checkpoint)

Epoch	Training Accuracy (%)	Validation Accuracy (%)
5	76.04	91.67
10	95.31	95.83
15	97.92	95.83
20	97.40	97.92
25	98.44	95.83
30	93.23	95.83
35	96.88	95.83
40	97.92	95.83
45	97.40	97.92
50	97.92	97.92

Loss Curve (Checkpoint)

Epoch	Training Loss	Validation Loss
5	0.548335	0.481803
10	0.307425	0.269742
15	0.157133	0.187459
20	0.121004	0.107578
25	0.090004	0.148434
30	0.213115	0.127387
35	0.106067	0.097069
40	0.077886	0.126727
45	0.085549	0.057880
50	0.069808	0.062094